



Microbial degradation of antibiotic: future possibility of mitigating antibiotic pollution

Aindrila Kayal · Sukhendu Mandal 

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Abstract Antibiotics are the major pharmaceutical wastes that are being exposed to the environment from the pharmaceutical industries and for the anthropogenic activities. The use of antibiotics for disease prevention and treatment in humans has been surpassed by the amount used in agriculture, particularly on livestock. It is stipulated that the overuse of antibiotics is the single largest reason behind the rise of bacterial anti-microbial resistance (AMR). The development of alternative therapy, like gene therapy, immunotherapy, use of natural products, and various nanoparticles, to control bacterial pathogens might be an alternative of antibiotics for mankind but the remediation of already exposed antibiotics from the lithosphere and hydrosphere needs to be envisioned with priority. The ever-increasing release of antibiotics in the environment makes it one of the major emerging contaminants (ECs). Decomposition of such antibiotic contaminants is a great challenge to get a cleaner environment. There are reports describing the degradation of antibiotics by photolysis, hydrolysis, using cathode and metal salts, or by degradation via microbes. Anti-microbials like sulfonamides are recalcitrant to natural biodegradation, exhibiting high thermal stability. There are recent reports on microbial degradation of a

few common antibiotics and their derivatives but their applications in waste management are scanty. It could however be a major concern to the scientists whether to use the antibiotic degradation traits of a microbe for the removal of antibiotic wastes. The complexity of the genetic clusters of a microbe that are responsible for degradation is crucial, as a small genetic cluster might have higher chance of horizontal transfer into sensitive species of the normal microbial flora that in turn triggers the rise of antimicrobial resistance.

Keywords Antibiotics · Antibiotic resistance · Pharmaceutical wastes · Waste management · Fungal remediation · Bacterial remediation

Abbreviations

ARGs	Antibiotic resistant genes
SM2	Sulfamethazine
SD	Sulfadiazine
SMX	Sulfamethoxazole
CIP	Ciprofloxacin
GEN	Gentamicin
NOR	Norfloxacin
OFX	Ofloxacin
WWTPs	Wastewater treatment plants

Introduction

The exposure of pharmaceutical wastes in environment increases over time (Kolpin et al., 2002). Antibiotics

A. Kayal · S. Mandal (✉)
Laboratory of Molecular Bacteriology, Department
of Microbiology, University of Calcutta, 35, Ballygunge
Circular Road, Kolkata 700019, India
e-mail: sukhendu1@hotmail.com

are one of the major pharmaceutical wastes that exert a potential threat to human civilization. Humans or animals can metabolize a small portion of antibiotics and the major unmetabolized remnants are being released into the environment (Nguyen et al., 2017). The hydrosphere also gets contaminated by antibiotics through direct or indirect human intervention like livestock breeding, sewage discharge, landfill, and by crop-land leachate. The presence of antibiotics in surface or groundwater has been estimated at levels of a few nanograms to milligrams per liter (Adelowo et al., 2009). Antibiotics are also available over-the-counter and people are impassive of using antibiotics following the standard guidelines and trashing the unused antibiotics into the environment. The omnipresence of such contaminating antibiotics influences normal bacterial flora to acquire antibiotic-resistant genes. The prevalence of anti-microbial-resistant (AMR) pathogens, by virtue of acquired antibiotic-resistant genes (ARGs), becomes obvious in almost all native microbiota. AMR pathogens of clinical relevance and their ARGs are more abundant in contaminated water like hospital wastewater and pharmaceutical industry wastewater. Human activities promote the spreading of the ARGs among microbes. In many cases, the ARGs are flanked by mobile genetic elements which trigger further spreading of ARGs across species. Therefore, first-generation antibiotics (for example, beta-lactams) have become inefficient to treat illnesses caused by a mild infection by the resistant pathogen. This practice will confer resistance to pathogenic as well as non-pathogenic beneficial microbes even against the newly discovered next-generation antibiotics in near future. Consequently, the bacterial infection will not be cured easily with available antibiotics and exert challenge to combat disease as it was during the pre-antibiotic era.

Recent studies have evidenced and clarified the degrading and transforming ability of environmental pollutants by microbes. Many wastewater treatment plants (WWTPs) use physicochemical pretreatment and activated sludge processes as a biological treatment for their sewage discharge containing less harmful effluent. In this report, we address the threat imposed by antibiotic pollution in environment that has never been in focus. Due to the increase in AMR, the antibiotics used so far for disease treatment are getting useless. As antibiotics cannot be degraded by the conventional waste management systems, their biological waste management could be a game-changer as it is easy

and cost-effective. In this review, we have discussed the potential of microbes for remediation of different classes of antibiotics exposed to the environment. Thus, the report can be very useful for the scientists in near future, and also this can be an alarm to consider antibiotic pollution seriously. Furthermore, the report overviews and illustrates the latent qualities of microbes to overcome antibiotic pollution.

Presence and fate of antibiotics in the environment

There is a continuous discharge of antibiotics after being used on livestock, humans, and crops. The use of antibiotics for disease cure, disease prevention, and growth promotion of livestock exponentially increases their exposure to the environment. Most of the industrial wastes are generally treated in wastewater treatment plants (WWTPs) and many intact or intermediate hazardous pollutants from the WWTPs may eventually be distributed in the surface water and finally have an impact on the water quality, the ecosystem, and human and animal health (Yuan et al., 2009). Sulfamethoxazole (SMX) is one of the widely used sulfonamides group of antibiotics which present in trace levels to $>200 \text{ mg kg}^{-1}$ in manure with an overall average of $1\text{--}10 \text{ mg kg}^{-1}$, which ultimately get discarded in high concentrations in the nature (Ramaswamy et al., 2010; Wang et al., 2018). Whereas, derivatives like Sulfathiazole and Sulfamethazine (SM2) have been detected in the manure at concentrations of 12.4 and 8.7 mg kg^{-1} , respectively (García-Galán et al., 2008). As high as 500 mg L^{-1} of Sulfadiazine (SD) was found to be present in the industrial sludge (Grote et al., 2004). A broad-spectrum antimicrobial agent Fluoroquinolones has been used widely as human and veterinary medicine (Brown, 1996). The most commonly used third-generation Fluoroquinolone is Ciprofloxacin (CIP), which has very slow biodegradability (Janecko et al., 2016). CIP is the second-highest Fluoroquinolone prescribed in China and was estimated about 5340 metric tons used in 2013 (Zhang et al., 2015b). CIP exposure results in a potential threat to the environment as it is hard to decompose even at high temperatures and has frequently been detected in manure composts (Pan et al., 2017). The concentration of CIP in stream water is about 0.01 to 0.03 mg L^{-1} (Maul et al., 2006) and about 0.75 mg kg^{-1} in agricultural soils (Martínez-Carballo et al., 2007). It is also detected at elevated

concentrations in pharmaceutical wastewater (6.5 to 31 mg L⁻¹) and hospital wastewater (10 to 200 µg L⁻¹) (Nguyen et al., 2017). CIP is found harmful for overall bio-geochemical cycles operating in the lithosphere as it regulates the adaptations and catabolic diversities of the microbial communities (Girardi et al., 2011; Ma et al., 2014). Gentamicin (GEN) is another commonly used broad-spectrum aminoglycoside antibiotic and is being exposed substantially into the environment during its production and application in hospitals and livestock farms (Adelowo et al., 2009).

Antibiotic waste management and its drawbacks

It has been reported that antibiotics can be degraded either by photolysis, hydrolysis, by cathode, or metal salts (Wang et al., 2016; Zhang et al., 2015a); however, antibiotics like Sulfonamides are stable at high temperature and are recalcitrant to biodegradation (Zhang et al., 2014) (Fig. 1).

Advanced oxidation processes (AOPs)

Ozone, Fenton, UV/hydrogen peroxide, and UV/per-sulfate mediated advanced oxidation processes showed higher efficacy of SMX degradation (Liu et al., 2017, 2018; Wang & Wang, 2016, 2017a, 2018a). AOP-mediated mineralization of SMX is very less which indicates the formation of intermediate products at the time of degradation of SMX. The complete mineralization of SMX can be achieved by higher chemical and energy input (Pérez et al., 2002).

Sorption

Sorption of Sulfadimethoxine, Sulfaquinoxaline, and Sulfamethazine was found to be higher in clay than in sandy soil for the intrinsic characters of the Sulfonamides and the organic content of soil (Doretto et al., 2014). Acidic pH can also increase the level of sorption of Sulfonamides (Zhang et al., 2014).

Hydrolysis

The degradation rate of Chlortetracycline and Tetracycline varies at different pH and temperatures. But the degradation of Lincomycin and Sulfonamides like

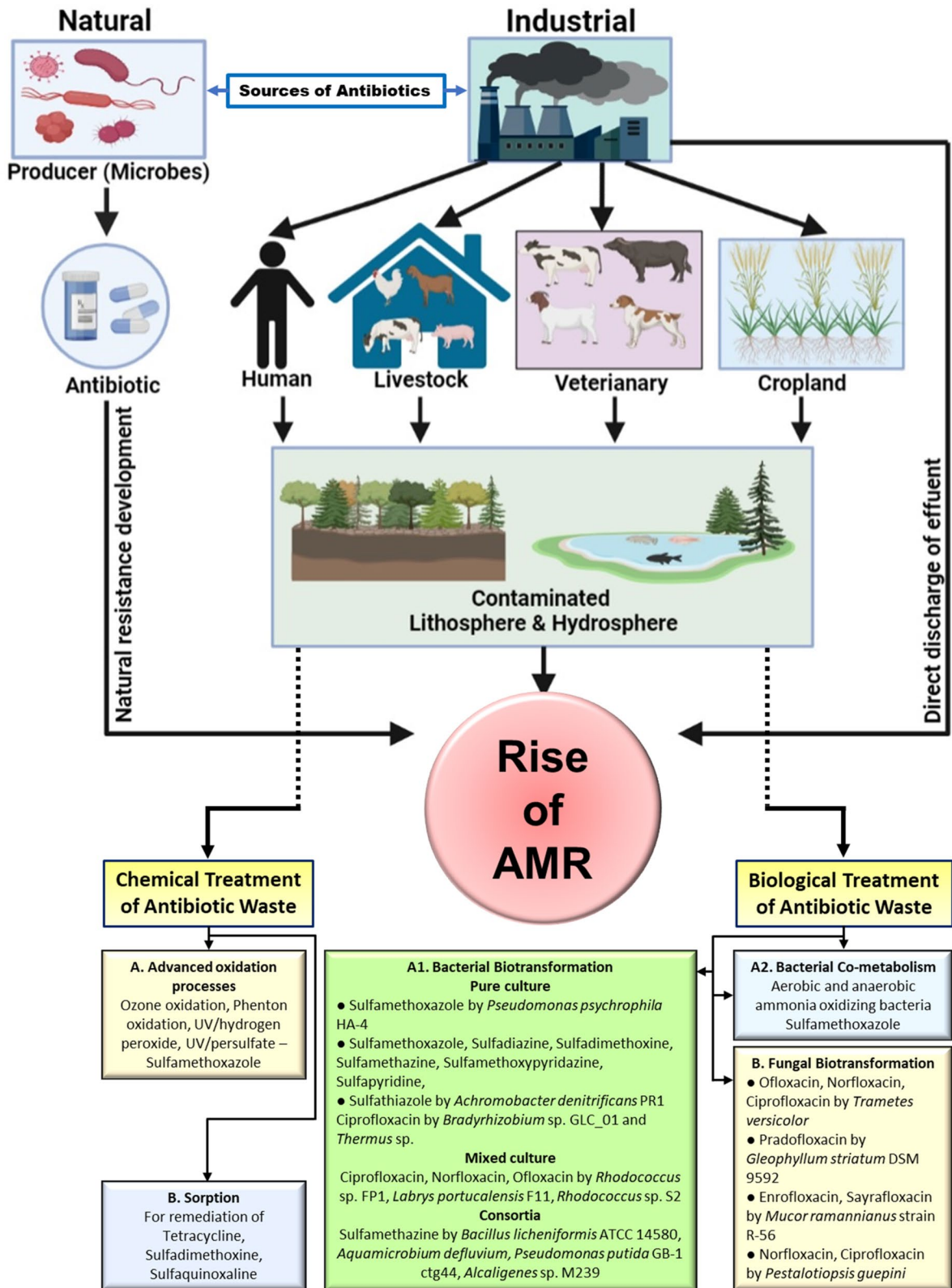
Sulfachlorpyridazine, Sulfadimethoxine, and Sulfathiazole is not affected significantly by the changes in pH or temperatures (Loffin et al., 2008).

Microbial remediation of antibiotics

Microbial degradation of pollutants mainly occurs either by their co-metabolism or utilization as growth substrates (Nzila, 2013).

Sulfonamides

Recently, an increasing number of Sulfonamides degrading bacteria have been isolated (Jiang et al., 2014; Reis et al., 2014). *Geobacillus thermoleovorans* strain S-07 can degrade Sulfamethazine (SM2) at high temperature. More than 95% of 10 mg L⁻¹ SM2 was reported to be degraded by the strain S-07 in 24 h at 70 °C and 6.0 pH (Ingerslev & Halling-Sørensen, 2000). The SMX-degrading *Achromobacter denitrificans* PR1 can also degrade Sulfonamides (Reis et al., 2014). However, the complete SMX degradation mechanisms are absent in *Microbacterium* sp. BR1, but in many reports, the 3-amino-5-methylisoxazole has constantly been identified as the main metabolic product during the SMX metabolism by this strain. Aminophenol and trihydroxybenzene were also found as the metabolic intermediates in SMX degradation by *Microbacterium* sp. BR1 (Ricken et al., 2013, 2017). 3-amino-5-methylisoxazole has also been found as the main metabolite in SMX degradation by *Pseudomonas psychrophila* HA-4 (Jiang et al., 2014). The SMX can be removed in lower rate from the fresh activated sludge, but removed completely from the acclimated activated sludge up to 160 mg L⁻¹ at 25 °C (Wang & Wang, 2017b). It is important to note that the SMX adsorption in aerobic sludge is very little; thus, biodegradation plays a major role in SMX removal (Lv et al., 2016). Both the aerobic and anaerobic ammonia-oxidizing bacteria are involved in ammonia oxidation (Lv et al., 2016). These bacteria can also remove SMX as high as 86% (100 µg L⁻¹ of initial concentration) when the waste material contains 1 g NH₄⁺ at 30 °C (Kassotaki et al., 2016). The SMX removal can further be increased by addition of acetate. Sulfate-reducing, iron-reducing, and methanogenic conditions help SMX removal by 80% (Zhang et al., 2013) which indicates that the bacteria having these properties may have SMX degradation ability. *Achromobacter denitrificans* effectively degrades both



◀**Fig. 1** Sources of antibiotic wastes and their degradation mechanisms. Natural and commercial sources of the antibiotic contaminate the lithosphere and hydrosphere and are the major triggering factor behind the rise of the antimicrobial resistance. The antibiotics present in the environment can be deactivated or degraded both by chemical treatment and by biological treatment

SMX and Sulfonamides (Reis et al., 2014). In wastewater treatment, the removal efficiency for SMX ranged from 0 to $> 98\%$ (Bendz et al., 2005; Peng et al., 2006).

Tetracyclines

A novel bacterium *Burkholderia cepacia* can degrade Oxytetracycline (OTC) up to $79.20 \pm 0.32\%$ and Tetracycline (TC) ($82.31 \pm 0.62\%$) (Hong et al., 2020). However, it has been shown in the past that the Tetracycline cannot be removed by biotransformation in a batch experiment (Kim et al., 2005).

Fluoroquinolones

Norfloxacin (NOR) insensitive *Microbacterium* sp. utilizes N-phenylpiperazine as carbon source. This bacterium cannot grow on NOR but can bio-transform it into four metabolites: 8-hydroxy NOR, 6-defluoro-6-hydroxy NOR, des-ethylene NOR, and N-acetyl-NOR (Kim et al., 2011). *Labrys portucalensis* F11 grows using fluorobenzene as a carbon and energy source and can able to co-metabolize Fluoroquinolone (Amorim et al., 2014). It can transform CIP, NOR, and Ofloxacin (OFX) (concentrations above $3.5 \mu\text{M}$) separately or from the mixture. Sixty percent OFX ($2 \mu\text{M}$) removal was observed when the cells were grown with $5.9 \mu\text{M}$ acetate supplementation, though the complete defluorination did not occur. Fluoroaromatic hydrocarbon-degrading strains *Rhodococcus* sp. FP1, *L. portucalensis* F11, and *Rhodococcus* sp. S2 transformed CIP in presence of acetate (Amorim et al., 2014; Maia et al., 2014). Eight *Microbacterium* strains were reported to transform NOR through the N-acetylation of the piperazine ring. The metabolites are less effective antibacterial compounds (Adjei et al., 2006). *Microbacterium* sp. 4N2-2 hydroxylated NOR in the C-8 position and modified its piperazine ring by N-acetylation or des-ethylation (Kim et al., 2011). The strain *L. portucalensis* F11 can eliminate up to $2 \mu\text{M}$ of NOR but only about 38% defluorination occurs (Amorim et al., 2014). *L. portucalensis* F11 can also transform up to $2 \mu\text{M}$ of OFX completely (Amorim et al., 2014).

However, the degradations of other fluoroquinolone compounds were found insufficient. *Bradyrhizobium* sp., isolated from activated sludge, can transform 45% of CIP in presence of glucose as carbon source (Nguyen et al., 2018). The biotransformation of CIP by the thermophilic bacterium, *Thermus* sp., has also been detected while growing in CIP-containing pharmaceutical sludge (Pan et al., 2018). Recalcitrant antibiotics, viz., tetracycline, trimethoprim, and CIP, are having poor biodegradability because they are unable to induce the expression of bacterial genes involved in the degradation. This describes why the pure culture cannot effectively degrade Tetracycline, Trimethoprim, and CIP (Wang et al., 2016).

Macrolides

It has also been reported that from the activated sludge system, up to 50% of Trimethoprim and Macrolides can be removed, and the removal was found to be enhanced up to 90% with increasing solid retention time (SRT) (Göbel et al., 2007). In another independent study, it has also been observed that the Roxithromycin, Clarithromycin, and Clindamycin can only be removed under anoxic conditions whereas Doxycycline was effectively removed under aerobic conditions (Burke et al., 2014).

Aminoglycosides

Gentamicin

Among the known GEN-degrading microbes, *Brevundimonas diminuta* BZC3 was found to have the highest degradation ability (50%) followed by the *Providencia vermicola*, *Alcaligenes* sp., and *Acinetobacter* sp. The acclimatized microflora in QD4 bio-solid sludge (AMQD4) consortia, consisting of *Providencia vermicola*, *Brevundimonas diminuta*, *Alcaligenes* sp., and *Acinetobacter*, can remove 56.8% and 47.7% GEN, respectively, from the unsterilized and sterilized sewage (Nzila, 2013). The highest removal efficiency of GEN (by AMQD4) was found at the lower GEN concentrations ranging from 50 to 100 mg L^{-1} (Liu et al., 2017).

Remediation using pure culture

Many pure bacterial cultures, isolated from activated sludge, have high SMX removal capacity. Some of these isolates can completely mineralize SMX by their

own (Wang & Wang, 2018b). Among these, *Microbacterium* sp. can completely remove 240 mg L⁻¹ of SMX and *Acinetobacter* sp. can remove 10 mg L⁻¹ of SMX. *Achromobacter denitrificans* PR1 could significantly enhance SMX removal if the medium is supplemented with additional carbon sources (Nguyen et al., 2017; Reis et al., 2014). This enhancement of SMX removal upon carbon addition was also observed in case of *Rhodococcus equi*. Presence of additional carbon source makes *Rhodococcus rhodochrous*, *Bacillus subtilis*, *Pseudomonas aeruginosa*, and *Alcaligenes faecalis* capable to degrade SMX (Nguyen et al., 2017; Reis et al., 2014). Pure cultures of these bacteria do not readily use antibiotics as carbon and energy source, but the added carbon and energy sources are necessary for their metabolic function. A pure strain, *L. portucalensis* F11, can transform 0.8 µM of CIP but the defluorination was not observed. It was reported before that a pure bacterial isolate hydroxylates and defluorinates fluoroquinolone (Table 1) (Kim et al., 2002, 2005, 2010, 2011). Likewise, a multi-drug-resistant strain, *Pseudomonas* sp. MR 02, isolated from the river water can metabolize Ampicillin as sole carbon and nitrogen and energy source (Ranjan et al., 2019).

Remediation using mixed culture

Antibiotic degradation by microbial consortia has been found more effective than that of the single microbe. After acclimation, a mixed microbial culture effectively utilizes SMX as their carbon and energy source (Wang & Wang, 2017b). At a steady state, during successive feedings of CIP, a mixed culture of *Rhodococcus* sp. FP1, *Rhodococcus* sp. S2, and *Labrys portucalensis* F11 defluorinated CIP, but at a slower rate than the overall elimination of CIP. When the concentration of CIP was increased, only an incomplete transformation took place where the substrate was partially defluorinated (22%) (Amorim et al., 2014). A mixed culture of *Rhodococcus* sp. S2, *Rhodococcus* sp. FP1, and *L. portucalensis* F11 was found to transform NOR and release 60.5% of the fluorine during the successive feedings in fluoride form (Maia et al., 2014). Likewise, the mixed culture of *Rhodococcus* sp. FP1, *L. portucalensis* F11, and *Rhodococcus* sp. S2 can transform Ofloxacin. At the end of 33 days, as high as 90% of the Ofloxacin was found to be bio-degraded (Maia et al., 2014). Moxifloxacin was reported to be degraded by a

mixed culture of *Rhodococcus* sp. FP1, *L. portucalensis* F11, and *Rhodococcus* sp. S2 at a lower extent than the degradation of fluoroquinolone by the pure cultures (Maia et al., 2014). Mixed culture of anaerobic sulfate-reducing bacteria (viz., *Desulfobacter* sp.) can degrade CIP by des-ethylation of the piperazine substituent and hydroxylation (Jia et al., 2018). Strains belonging to genera *Dysgonomonas*, *Ferruginibacter*, *Leucobacter*, *Pseudoxanthomonas*, *Phenylobacterium*, and *Stenotrophomonas* were predominated among the CIP degrading bacterial community (Table 1) (Liao et al., 2016).

Remediation using fungal strains

Eibes et al. (2011) reported that the Sulfamethoxazole can be degraded efficiently (80%) by *Bjerkandera adusta* (ATCC 90,940) with the help of its ligninolytic peroxidase. *Gleophyllum striatum* DSM 9592 also has been reported to degrade the Pradofloxacin and catechol-type metabolite, and at a late-stage of growth, an opened-ring metabolite was formed. Complete degradation of this class of antibiotics was demonstrated for the first time by Wetzstein et al. (2012). *Mucor ramannianus* R-56 has been reported to bio-transform Enrofloxacin where the bio-transformed products were des-ethylene-enrofloxacin, enrofloxacin-N oxide, and N-acetyl-ciprofloxacin. The same fungal strain was also found to transform Sarafloxacin (veterinary antibiotic) and produce N-acetylsarafloxacin (26%) and desethylene-N-acetylsarafloxacin (15%) (Parshikov et al., 2000). Another fungal strain *Trametes versicolor* can also transform as high as 80% of the Ofloxacin after 7 days of incubation (Gros et al., 2014). The strain of *T. versicolor* and *Pestalotiopsis guepini* also reported to bio-transform Norfloxacin and Ciprofloxacin (Table 2) (Parshikov et al., 2001b; Prieto et al., 2011).

Advantages of biological remediation over other conventional processes

Biological treatment of environmental pollutants is advantageous over the chemical treatment for being environmentally friendly and cost-effective. Application of microbial degradation and bioremediation technology is increasing for the removal of antibiotics from the environment (Liu et al., 2017). However, biological

Table 1 List of bacterial strains highlighting their potential attributes in degradation of antimicrobial compounds

Bacterial strain	Degradation product	Remarks	Reference
Pure culture			
<i>Sulfamethazine (SM2), Sulfamerazine (SM1), Sulfamethoxazole (SMX), Sulfadiazine (SDZ)</i>			
<i>Geobacillus thermoleovorans</i>	95% SM2 removed in 24 h. 85.96% SM1 removed in 48 h	Complete removal was not observed	Pan et al. (2017)
<i>Microbacterium</i> sp. strain BR1	Main degraded product- 3-amino-5-methylisoxazole	Complete degradation of SMX SDZ, Sulfadimethoxine, SM2, Sulfamethiazole degradation rate is 2.51 $\mu\text{M}/\text{min}/\text{g}$, 1.64 $\mu\text{M}/\text{min}/\text{g}$, 1.53 $\mu\text{M}/\text{min}/\text{g}$, 2.26 $\mu\text{M}/\text{min}/\text{g}$, respectively	Ricken et al. (2013)
<i>Achromobacter denitrificans</i> PRI	ND	SMX removal rate is $73.6 \pm 9.6 \mu\text{mol}/\text{h}/\text{g}$ cell. SD, Sulfadimethoxine, SM2, Sulphamethoxypyridazine, Sulfapyridine, Sulfathiazole degradation in 56 h is $98 \pm 2\%$, $48 \pm 1\%$, $100 \pm 0\%$, $98 \pm 2\%$, $100 \pm 0\%$, $47 \pm 13\%$, respectively	Reis et al. (2014)
<i>Pseudomonas psychrophila</i> HA-4	Major intermediates: 3-amino-5-methylisoxazole. Aniline, 4-aminothiophenol, and Sulfanilamide	34.30% SMX removed after 192 h at 10 °C	Jiang et al. (2014)
Oxytetracycline (OTC), Tetracycline (TC)			
<i>Burkholderia cepacia</i> OTC-1	ND	Degradation efficiency of TC and OTC is $82.31 \pm 0.62\%$ (degradation rate of $2.26 \text{ mg kg}^{-1} \text{ d}^{-1}$) and $79.20 \pm 0.32\%$	Hong et al. (2020)
Ciprofloxacin (CIP), Ofloxacin (OFX), Norfloxacin (NOR)			
<i>Microbacterium</i> sp. 4N2-2	Biotransformation products are 8-hydroxy NOR, 6-defluoro-6-hydroxy NOR, des-ethylene NOR, and N-acetyl-NOR (N-acetylation occurs here)	End metabolites subsequently less toxic compared to NOR	Kim et al. (2011)
<i>Labrys portucalensis</i> F11	CIP-A and CIP-B are from CIP; OFX-A and OFX-B are from OFX; and NOR-A, NOR-B, and NOR-C are from NOR	Intermediates less toxic than the parent compounds	Amorim et al. (2014)
<i>Bradyrhizobium</i> sp. GLC_01	7-amino-1-cyclopropyl-6-fluoro-4-oxo-1,2,3,4-tetrahydroquinoline-3-carboxylic acid and 1-cyclopropyl-6-fluoro-8-hydroxy-4-oxo-7-(piperazinyl-1-332 y1)-1,2,3,4-tetrahydroquinoline-3-carboxylic acid)	Biotransformation of over 70% CIP at 0.05 mg L^{-1}	Nguyen et al. (2018)
<i>Mycobacterium</i> sp. PYR100 <i>M. gilvum</i> PYRGCK	N-acetyl CIP, N-nitroso CIP	Reduced antibacterial activity of CIP	Adjei et al. (2007)

Table 1 (continued)

Bacterial strain	Degradation product	Remarks	Reference
<i>Lactobacillus gasseri</i>	NA	Complete removal of CIP	Liyang and Manage (2018)
<i>Enterobacter</i> sp.	NA	96% degradation observed	
<i>Bacillus</i> sp.	NA	74% degradation observed	
<i>B. subtilis</i>	NA	70% degradation observed	
<i>M. luteus</i>	NA	56% degradation observed	
<i>Thermus</i> sp.	NA	Degraded 52% of 5 mg L ⁻¹ CIP	Pan et al. (2018)
Mixed bacterial consortia			
Sulfate-reducing bacteria	NA	84.2% SMX removed	Jia et al. (2017)
<i>Burkholderia cepacia</i> , <i>P. nitroreducens</i>	NA	85.31 ± 0.25% degradation	Hong et al. (2020)
<i>Rhodococcus</i> sp. FP1, <i>Labrys portucalensis</i>	CIP-A, CIP-B, CIP-C, CIP-D, CIP-E, and CIP-F are from CIP; NOR-A, NOR-B, and NOR-C are from NOR; OFX-A, OFX-B, and OFX-C are from OFX	95.6% CIP degraded by 33 days, 96.1% NOR degraded by 19 days, 90% OFX degraded at the end 33 days	Maia et al. (2014)
<i>Desulfobacter</i> sp.	ND	CIP—effectively biodegraded (61.0 ~ 637.3 µg/g-SS/d)	Jia et al. (2018)
<i>B. licheniformis</i> ATCC 14580, <i>Aquamicrobium defluvium</i> , <i>P. putida</i> GB-1 ctg44, <i>Alcaligenes</i> sp. M239	ND	SMZ—consortia remove up to 7.8% of after 20 days	Islas-Espinoza et al. (2012)
Gentamicin (GEN)			
AMQD4 (<i>Providencia vermicola</i> , <i>Brevundimonas diminuta</i> , <i>Alcaligenes</i> sp., and <i>Acinetobacter</i>)	Main degradation product was 3'-acetylgentamicin	Addition of AMQD4 in unsterilized sewage promoted the degradation of gentamicin	Liu et al. (2017)

Table 2 List of fungal strains highlighting their potential attributes in degradation of antimicrobial compounds

Strain	Parent and degradation product	Remarks	Reference
<i>Bjerkandera adusta</i> ATCC90940	Degradation of SMX	80% SMX removed	Eibes et al. (2011)
<i>Gleophyllum striatum</i> DSM 9592	Pradofloxacin (PRA)–catechol-type metabolite and late-stage opened ring metabolite	Complete degradation of PRA demonstrated first-time	Wetzstein et al. (2012)
<i>Mucor ramannianus</i> R-56	Enrofloxacin (ENR)–biotransformed products are desethylene-ENR, ENR-N oxide, N-acetyl-CIP	Less toxic than ENR	Parshikov et al. (2000)
<i>Trametes versicolor</i>	OFX–80% OFX degraded after 7 days of incubation	No defluorinated intermediates found	Gros et al. (2014)
<i>Mucor ramannianus</i> R-56	Sarafloxacin (SF)–two metabolites: N-acetyl-SF (26%), Desethylene-N-acetyl-SF (15%)	Less toxic than SF	Parshikov et al. (2001a)
<i>Gleophyllum striatum</i> DSM 9592	PRA–six major metabolites are formed	Derived product loses antibacterial activity	Wetzstein et al. (2012)
<i>Pestalotiopsis guepini</i>	NOR–metabolites are N-acetyl-NOR (55.4%), desethylene-N-acetyl-NOR (8.8%), N-formyl-NOR (3.6%), and 7-amino-1-ethyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid (2.1%)	No fluorinated intermediate found	Parshikov et al. (2001b)
<i>Trametes versicolor</i>	NOR–transformation products are 7-((2-aminoethyl) amino)-1-ethyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid, 7-Amino-1-ethyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid, 7-((2-Acetamidoethyl) amino)-1-ethyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid	NA	Prieto et al. (2011)
<i>Trametes versicolor</i>	CIP–biotransformed products are CIP-1, CIP-2, CIP-3, CIP-4, CIP-5, and CIP-6. Those are 7-((2-aminoethyl) amino)-1-cyclopropyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid, 7-amino-1-cyclopropyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid, 1-cyclopropyl-6-fluoro-8-hydroxy-4-oxo-7-(piperazin-1-yl)-1,4-dihydroquinoline-3-carboxylic acid, 7-((2-acetamidoethyl) amino)-1-cyclopropyl-6-fluoro-8-hydroxy-4-oxo-1,4-dihydroquinoline-3-carboxylic acid	NA	Prieto et al. (2011)
<i>Pestalotiopsis guepini</i>	CIP–metabolites are N-acetyl-CIP (52.0%), desethylene-N-acetyl-CIP (9.2%), N-formyl-CIP (4.2%), and 7-amino-1-cyclopropyl-6-fluoro-4-oxo-1,4-dihydroquinoline-3-carboxylic acid (2.3%)	No defluorination observed	Parshikov et al. (2001b)
<i>Pleurotus ostreatus</i> <i>P. kumm</i> SMR684	OTC–a putative degradation product, ADOTC (2-acetyl-2-decarboxamido-OTC) is proposed	This fungus can almost completely degrade OTC in few days. The drug is absorbed by its mycelia	Migliore et al. (2012)

treatment shows inconsistent antibiotic-removal capacity. The activated sludge, the most conventional step in WWTPs, does not have an obvious effect on the removal of SMX (Blair et al., 2015). However, about 90% of SMX can be removed through the acclimated activated sludge (Wang & Wang, 2017b). There are concerns to remove certain heat and acidic or alkaline-resistant antibiotics (like Gentamicin) through chemical or physical methods. *Trametes versicolor* alone can remove CIP and NOR more than 90% (Selvi et al., 2014). A consortium that consists of *Aquamicrobium defluvium*, *Alcaligenes* sp., *Bacillus licheniformis*, and *Pseudomonas putida* is able to enhance the degradation of sulfonamides from soil (Islas-Espinoza et al., 2012). Again, survival or consistently maintaining high degradation efficiency in alkaline or acidic pH by the AMQD4 consortia (acclimatized microflora in QD4 bio-solid sludge) is an important parameter for future waste management of the GEN production industries (Liu et al., 2017). Formulation of independent consortia for the bioremediation of each antibiotic would be a better choice for future antibiotic or pharmaceutical waste management.

Impact of antibiotic biodegradation on environment

Selection of microbes having the efficiency to degrade antibiotics is a cumbersome process. Most often, the segregation of antibiotic-degrader from the resistant bacteria suffers from false positive result. In laboratories with poor infrastructural facility, scientist may interpret the antibiotic-resistant autotrophic bacteria as an antibiotic-degrading one. Even if a bacterium is chosen based on the presence of specific gene that codes an enzyme (for example, blaNDM1 gene encoding New Delhi metallo- β -lactamase) capable of degrading an antibiotic, it might not be useful for environmental bioremediation as there lies a chance to disseminate the ARGs to the normal microflora. Thus, the environment gets further contaminated due to unwanted passing of ARGs from the donor to the recipient bacteria. However, the bacteria having a complex genetic cluster responsible for degradation pathway could be a fair choice for mitigation of the antibiotic pollution from lithosphere as well as hydrosphere as there are very low chances of getting the antibiotic-degrading gene clusters to the commensal

present in the same environment. Hence, the selection of bacteria for degradation of antimicrobials is a critical step for having possible deleterious impact on the environment.

Conclusion and future perspectives

The ever-increasing amount of antibiotics loaded in the lithosphere and hydrosphere is a major concern of the environmentalist. Steps for mitigating the antibiotics from the environment are a great challenge in the focus of the rise of deadly antimicrobial resistance. Further research is needed to get an idea of whether pure culture or mixed or consortia can successfully transform these highly toxic pharmaceutical compounds into less toxic ones. Also, it must be crucial to identify the molecular mechanisms of each degradation pathway by the individual organisms. Several measures like biological acclimation and bio-augmentation are adapted to increase the efficacy of the treatment through enriching the microbes associated with degradation. More efforts should be given to isolate antibiotic-degrading bacteria and also to identify and optimize their normal growth requirements. It could however be a major concern to the scientists whether to use the antibiotic degradation traits of a microbe for the removal of antibiotic wastes. The complexity of the genetic clusters of a microbe that are responsible for degradation is decisive, as a small genetic cluster might have higher chance to horizontally transfer to normal microbial flora and make the antimicrobial resistance or dissemination of antibiotic-resistant genes (ARGs) even worse. Thus, the genes or pathways of antibiotic inactivation or degradation might magnify among other recipient bacteria to turn the good strategy bad. Possibly this is the single reason that led to abandon the global research on biodegradation of antibiotics. Thus, in the era of advance biotechnology, there should be further research on biodegradation of such emerging pollutants with a deeper understanding of the multi-gene clusters responsible for the degradation pathway. Future research on microbial antibiotic degradation might open up new avenues to resolve the limitations for successful antimicrobial waste management.

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Declarations

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